# Interpreting patterns

Module 4 - Lesson 3

## Overview

A model’s patterns should make sense. Interpretation methods reveal which features drive predictions, so you can validate, explain, and trust the model.

## Key Concepts

- Feature importance (from tree models) ranks inputs by their contribution.
- Coefficients in linear models give direction and magnitude of effect.
- Partial dependence plots show how a feature changes predictions on average.
- SHAP values explain individual predictions and feature contributions.
- Interpretation catches data bugs and unintended shortcuts.
- Explain to stakeholders in plain language, not just numbers.

## Example

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.inspection import permutation_importance

model.fit(X_train, y_train)
perm = permutation_importance(model, X_test, y_test, n_repeats=10, random_state=42)
import numpy as np
for i in np.argsort(perm.importances_mean)[::-1]:
    print(X.columns[i], round(perm.importances_mean[i], 4))
```

## Summary

Interpretation turns a black box into an explainable decision. Feature importance, coefficients, and SHAP reveal drivers and catch bugs before trusting predictions.

## Practice Checklist

- Rank features by importance for a trained model.
- Generate SHAP values for one prediction and explain it.
- Write a plain-language explanation of what the model relies on.